MECHANISM OF ACTION

Understanding the pharmacology of olanzapine

A broad receptor-binding profile contributes to its pharmacological effects.

Olanzapine is an antipsychotic, antimanic and mood-stabilising agent with activity across several neurotransmitter receptor systems.[1]

Receptor profile | Receptor occupancy | Clinical relevance

Receptor-binding profile

DESCRIBED IN RELATION TO ANTIPSYCHOTIC ACTIVITY

Dopamine receptor activity

D1, D2, D3, D4, D5

Olanzapine has affinity for dopamine receptors, including D2 receptors.[1]

Dopamine receptor antagonism is relevant to its antipsychotic activity. Dopaminergic signalling is involved in multiple brain functions, and the relationship between receptor activity and individual symptom domains is complex.

DESCRIBED IN RELATION TO ANTIPSYCHOTIC ACTIVITY

Serotonin receptor activity

5-HT2A/2C, 5-HT3, 5-HT6

Olanzapine also has affinity for serotonin receptors, including 5-HT2A and 5-HT2C receptors.[1]

Its serotonin and dopamine receptor activity contributes to its overall pharmacological profile. However, the clinical effects of olanzapine cannot be attributed to one receptor interaction alone.

DESCRIBED IN RELATION TO ADVERSE EFFECTS

Histamine H1 receptors

H1

Relevant to effects such as somnolence and increased appetite.[13]

DESCRIBED IN RELATION TO ADVERSE EFFECTS

Muscarinic receptors

M1–M5

Relevant to anticholinergic adverse effects.[13]

DESCRIBED IN RELATION TO ADVERSE EFFECTS

Alpha-1 adrenergic receptors

α1

Relevant to orthostatic hypotension and dizziness.[13]

Receptor families at which olanzapine showed affinity (Ki < 100 nM) in preclinical studies. Relative affinities are not represented. These interactions help explain why treatment selection and monitoring must consider both therapeutic objectives and potential adverse effects.[1]

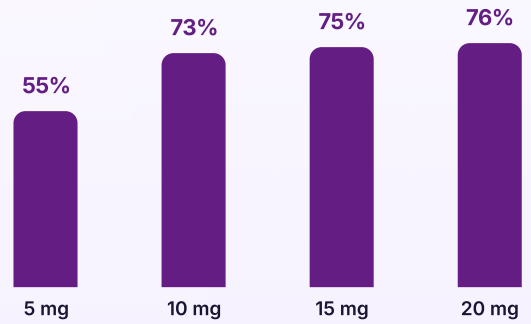
5-HT2 and D2 occupancy across the dose range

Olanzapine has greater affinity for serotonin 5-HT2 than dopamine D2 receptors. In a PET study in patients with schizophrenia, 5-HT2 occupancy was above 90% at every dose, while D2 occupancy rose with dose.[1,12]

Table. Striatal D2 and 5-HT2 occupancy across daily d

Dose	Mean D2	Individual D2 range	5
5 mg/day	**55%**	43–64%	>
10 mg/day	**73%**	71–75%	>
15 mg/day	**75%**	73–76%	>
20 mg/day	**76%**	74–80%	>

Three patients per dose, at steady state.[12]



Mean D2 occupancy by dose. 5-HT2 occupancy was above 90% at every dose from 5 to 20 mg/day (individual values about 91% to 100%). Doses above 20 mg/day reported in the paper are not shown.[12]

Study design and limitations

Twelve patients with schizophrenia were randomly assigned to fixed daily doses of 5–20 mg. Striatal D2 occupancy was measured with [11C]raclopride PET; baseline was estimated from age-corrected controls rather than each patient's own scan.[12]

Receptor occupancy does not by itself predict the clinical response of an individual patient.

Individual occupancy in 12 patients; numbers show mean D2 occupancy per dose. Redrawn from Kapur et al. 1998, Table 1. Doses above 20 mg/day reported in the paper are not shown.[12]

From pharmacology to clinical practice

A mechanism-of-action explanation provides context for understanding the pharmacology of olanzapine. It does not, by itself, establish the magnitude of clinical benefit or predict the response of an individual patient.

Clinical efficacy and safety should be judged on the clinical evidence and the prescribing information.

Review the clinical evidence →

Safety and prescribing

← Previous screen: **Disease background**

Next screen: **Dosing** →

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